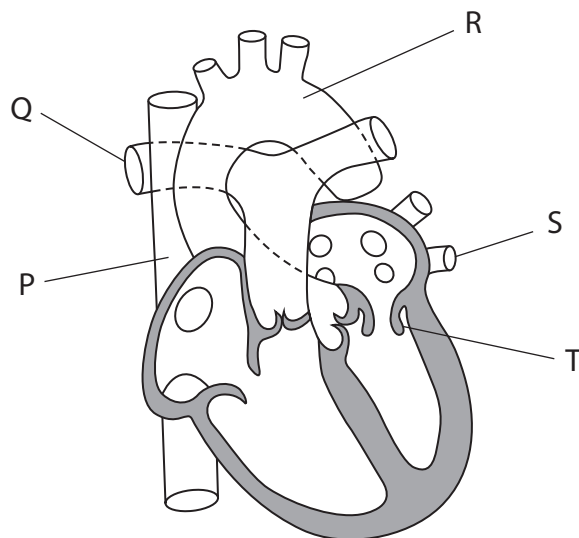


4 (a) The diagram shows the internal structure of a mammalian heart.



(i) How many of the following are correct for blood vessel P?

(1)

- carries blood to the heart
- has valves along its length
- has an endothelial lining

- ☐ **A** 0
- ☐ **B** 1
- ☐ **C** 2
- ☐ **D** 3

(ii) Why is blood vessel Q classified as an artery?

(1)

- ☐ **A** because it carries deoxygenated blood
- ☐ **B** because it carries oxygenated blood
- ☐ **C** because it carries blood away from the heart
- ☐ **D** because it carries blood back to the heart



(iii) Which blood vessel does the coronary artery branch from?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(iv) When in the cardiac cycle is valve T open?

(1)

- ☐ **A** atrial systole and cardiac diastole
- ☐ **B** atrial systole only
- ☐ **C** cardiac diastole and ventricular systole
- ☐ **D** ventricular systole only

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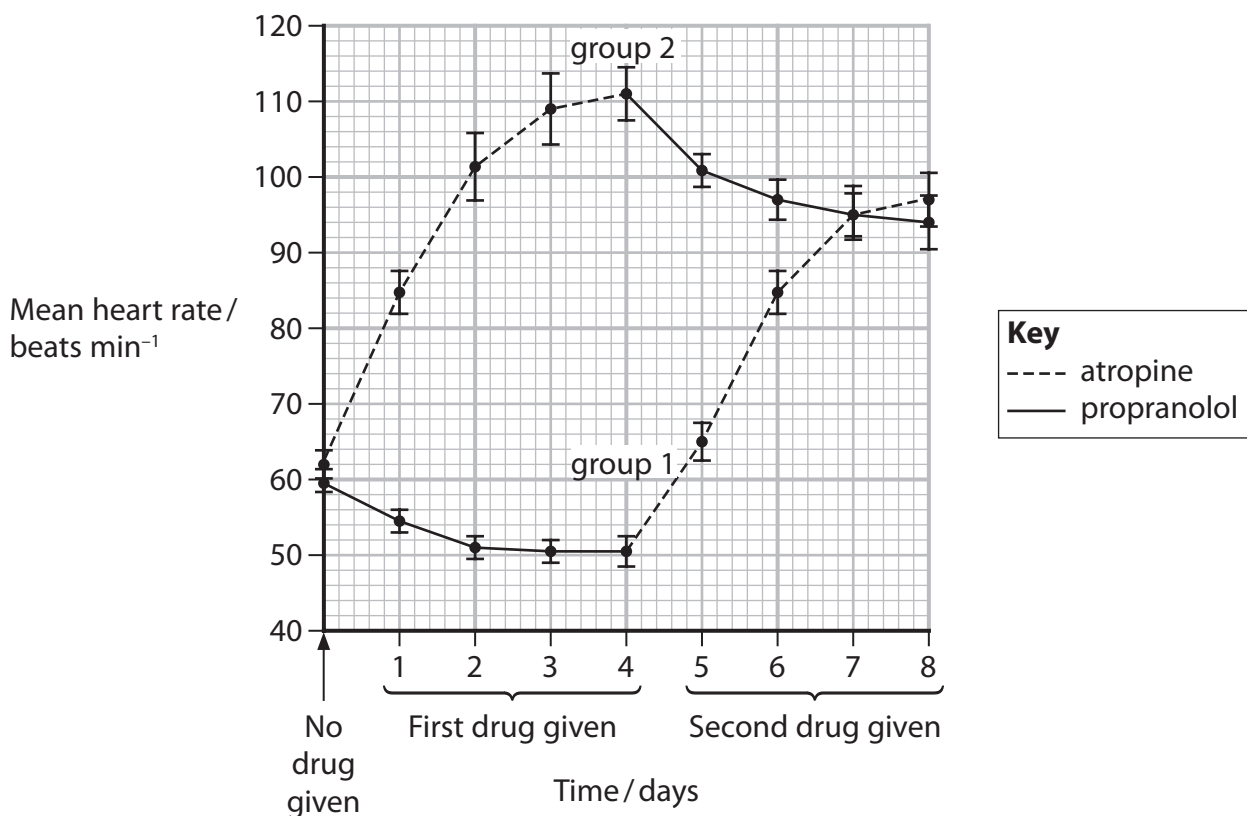


(b) The effect of atropine and propranolol on heart rate was investigated.

One group of people, group 1, were given propranolol each day for four days followed by atropine for four days.

A second group of people, group 2, were given atropine each day for four days followed by propranolol for four days.

The graph shows the results of this investigation.



- (i) Calculate the change in mean heart rate for group 2 after 8 days.

(1)

Answer beats min^{-1}

- (ii) Calculate the mean length of the cardiac cycle for group 2 on day 4.

(1)

Answer seconds

- (iii) Comment on the conclusions that can be drawn from this investigation.

(3)

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(Total for Question 4 = 9 marks)

